Reinforcement Learning With Reward Machines in Stochastic Games

Reinforcement Learning Course - Final Project — 06-2026

Matteo Liotta
matteo.liotta@studenti.units.it

Mattia Simonutti
mattia.simonutti@studenti.units.it

University of Trieste
Department of Mathematics and Geosciences

Problem Setting

- **Problem:** How to tackle complex tasks in **multi-agent RL** where reward functions are **non-Markovian**?
- **Proposed Approach:** Usage of **Reward Machines** to incorporate the high-level knowledge of the task. By RM state incorporation, form an augmented product stochastic game, **converting non-Markovian rewards** into **standard Markovian ones** [1].
- **Goal:** Learn and converge to the optimal best-response strategy at a **Nash equilibrium** for each agent in a **general-sum game setting**.

Basic Definitions

Stochastic Games with Non-Markovian Rewards

Definition — Two-Player General-Sum Stochastic Game (SG)

Modeled as the tuple $\mathcal{G} = (S, s_I, A_{\mathfrak{e}}, A_{\mathfrak{a}}, p, R_{\mathfrak{e}}, R_{\mathfrak{a}}, \gamma)$, where:

- **Finite State Space (S):** consisting of states $s = [s_{\mathfrak{e}}; s_{\mathfrak{a}}]$, with $s_{\mathfrak{e}}, s_{\mathfrak{a}}$ substates for *Ego* and *Adv* agents respectively. $s_I \in S$ is the initial state.
- **Action Spaces:** Each agent has its own finite set of actions $A_{\mathfrak{e}}$ and $A_{\mathfrak{a}}$.
- **Probabilistic Transition Function**: $p : S \times A_{\mathfrak{e}} \times A_{\mathfrak{a}} \times S \to [0, 1]$
- **Non*-Markovian Reward functions**: $R_{\mathfrak{e}\ or\ a} : (S \times A_{\mathfrak{e}} \times A_{\mathfrak{a}})^+ \times S \to \mathbb{R}$ map from **complete trajectory histories** to real value domains, specifying the n-m rewards for both agents.

(*) Usual definition assumes system Markovianity.

Trajectories and Rewards

- **Trajectory:** Sequence of states and actions $t_{0:k+1} = s_0 a_{\mathfrak{e},0} a_{\mathfrak{a},0} s_1 ... s_k a_{\mathfrak{e},k} a_{\mathfrak{a},k} s_{k+1}$ with $s_0 = s_I$.
- **Reward Sequence:** Given a **trajectory**, each agent ($i \in \{\mathfrak{e}, \mathfrak{a}\}$) receives reward sequence $r_{i,1} ... r_{i,k}$, where $r_{i,k} = R(s_0, a_{\mathfrak{e},0}, a_{\mathfrak{a},0}, ..., s_k)$.
- **Shared Observation:** Both agents **observe the same** trajectory.
- **Discounted Cumulative Reward:** Given a **trajectory** $t_{0:k+1}$, agents achieves a DCR reward $\sum_{t=0}^{k} \gamma^t r_{\mathfrak{i},t}$
- **Finite episodes assumption**: The game is episodic with a finite horizon, ensuring that the cumulative reward is well-defined.

Strategies and Objectives

- **Markovian Strategy:** Agents are equipped with a $\pi_i : S \times A_i \to [0, 1] \quad i \in \{\mathfrak{e}, \mathfrak{a}\}$ strategy, defining action given state selection probability.
- **Objective:** Maximize the **expected** discounted cumulative reward given policy of both agents, from both point of view.

SG Expected Discounted Cumulative Reward (Value Function)

Given agent $i \in \{\mathfrak{e}, \mathfrak{a}\}$, the expected DCR under policies $\pi_{\mathfrak{e}}$ and $\pi_{\mathfrak{a}}$ is defined as:

$$\tilde{v}_i(s, \pi_{\mathfrak{e}}, \pi_{\mathfrak{a}}) = \sum_{t=0}^{\infty} \gamma^t \mathbb{E}(r_{i,t} | \pi_{\mathfrak{e}}, \pi_{\mathfrak{a}}, s_0 = s)$$

Nash Equilibrium

Each agent's strategy is a **best-response to the other agent's strategy**, if each agent **cannot improve its own reward** by changing its strategy, given a **fixed opponent's policy**.

Definition: Nash Equilibrium in Stochastic Games

For a stochastic game $\mathcal{G} = (S, s_I, A_e, A_a, R_e, R_a, \gamma, p)$, the strategies of the ego agent, π_e^*, and the adversarial agent, π_a^*, are at a **Nash equilibrium** if:

$$\tilde{v}_e(s, \pi_e^*, \pi_a^*) \geq \tilde{v}_e(s, \pi_e, \pi_a^*)$$
$$\tilde{v}_a(s, \pi_e^*, \pi_a^*) \geq \tilde{v}_a(s, \pi_e^*, \pi_a)$$

hold for any π_e and π_a.

Objectives (cont'd)

The **actual objective** of each agent is to find the policy π_i^*, maximising the **expected discounted cumulative reward** considering the other agent's action once it reaches the game's Nash equilibrium

Agent Objectives at Nash Equilibrium

$$\pi_{\mathfrak{e}}^* = \arg\max_{\pi_{\mathfrak{e}}} \tilde{v}_i(s, \pi_{\mathfrak{e}}, \pi_{\mathfrak{a}}^*)$$

$$\pi_{\mathfrak{a}}^* = \arg\max_{\pi_{\mathfrak{a}}} \tilde{v}_i(s, \pi_{\mathfrak{e}}^*, \pi_{\mathfrak{a}})$$

Reward Machines (RM)

Definition: Reward Machine

Finite state automaton designed to encode non-Markovian rewards with state transitions and output functions: $\mathcal{A} = (V, v_I, 2^{\mathcal{P}}, \mathbb{R}, \delta, \sigma)$

- **Input Alphabet ($2^{\mathcal{P}}$):** Interprets propositional logic variables $\mathcal{P}$ generated from lower-level environment updates.
- **Set of RM states (V):** Finite collection of internal tracking states, with v_I as the initial state.
- **Transition function:** Deterministic state progression $\delta : V \times 2^{\mathcal{P}} \to V$.
- **Output function:** Discrete scalars along active trajectory $\sigma : V \times 2^{\mathcal{P}} \to \mathbb{R}$.

Game Theory Setting

Reward Machine Encoding

Definition 4: Reward Machine Encoding

A reward machine $\mathcal{A}_i = (V_i, v_{I,i}, 2^{\mathcal{P}}, \mathbb{R}_i, \delta_i, \sigma_i)$ encodes the non-Markovian reward function R_i of agent i in the stochastic game $\mathcal{G}$ if for every trajectory and corresponding event sequence $l_0 l_1 \dots l_k$, it holds that:

$$r_{i,0} \dots r_{i,k} = \mathcal{A}_i(l_0 l_1 \dots l_k)$$

- Formalizes RMs as high-level task specifications.
- Each agent uses a separate RM $(\mathcal{A}_{\mathfrak{e}}, \mathcal{A}_{\mathfrak{a}})$ to track independent non-Markovian goals.

The Augmented State Space Framework

Labeling Function (L)

Maps environment transitions to logical propositions:

$$L : S \times A_{\mathfrak{e}} \times A_{\mathfrak{a}} \times S \to 2^{\mathcal{P}}$$

$l_t = L(s_t, a_{\mathfrak{a},t}, a_{\mathfrak{e},t}, s_{t+1})$ is the set of true propositions in $\mathcal{P}$ at step t.

Lemma 1: Markovian Restoration

The product stochastic game has **Markovian reward functions** that express the original non-Markovian rewards.

- Given l_t, the RM state transitions to $v_{t+1} = \delta(v_t, l_t)$ and outputs reward $r_t = \sigma(v_t, l_t)$.
- This resolves historical dependencies and enables standard Q-learning.

Product Stochastic Game

Definition 5: Product Stochastic Game

Given an SG $\mathcal{G}$ and RMs $\mathcal{A}_{\mathfrak{e}}, \mathcal{A}_{\mathfrak{a}}$, the product SG is:

$$\mathcal{H} = (S', s'_I, A_{\mathfrak{e}}, A_{\mathfrak{a}}, p', R'_{\mathfrak{e}}, R'_{\mathfrak{a}}, \gamma)$$

where:

- **Augmented State Space:** $S' = S \times V_{\mathfrak{e}} \times V_{\mathfrak{a}}$. An augmented state is $s' = (s, v_{\mathfrak{e}}, v_{\mathfrak{a}})$.
- **Initial Augmented State:** $s'_I = (s_I, v_{I,\mathfrak{e}}, v_{I,\mathfrak{a}})$.
- **Transition Probabilities:** $p'(s', a_{\mathfrak{e}}, a_{\mathfrak{a}}, \bar{s}') = p(s, a_{\mathfrak{e}}, a_{\mathfrak{a}}, \bar{s})$ if $\bar{v}_i = \delta_i(v_i, L(s, a_{\mathfrak{e}}, a_{\mathfrak{a}}, \bar{s}))$ for $i \in \{\mathfrak{e}, \mathfrak{a}\}$, and 0 otherwise.
- **Markovian Reward Functions:**

$$R'_i(s', a_{\mathfrak{e}}, a_{\mathfrak{a}}, \bar{s}') = \sigma_i(v_i, L(s, a_{\mathfrak{e}}, a_{\mathfrak{a}}, \bar{s}))$$

The Stage Game Matrix

Definition: Stage Game

A two-player stage game is defined as $(r_{\mathfrak{e}}, r_{\mathfrak{a}})$, where r_i is agent i's reward function R_i, over the space of joint actions:

$$r_i = \{R_i(a_{\mathfrak{a}}, a_{\mathfrak{e}}) | a_{\mathfrak{a}} \in A_{\mathfrak{a}}, a_{\mathfrak{e}} \in A_{\mathfrak{e}}\}$$

- In QRM-SG, a stage game is formulated at each time step based on current Q-functions in augmented state space.
- The **Lemke-Howson method** is used to derive best-response strategies at a Nash equilibrium.

Nash Equilibrium & The Stage Game

Augmented Q-Function at Nash Equilibrium

Defined over the coupled augmented state space $S \times V_{\mathfrak{e}} \times V_{\mathfrak{a}}$:

$$q_i^*(s, v_{\mathfrak{e}}, v_{\mathfrak{a}}, a_{\mathfrak{e}}, a_{\mathfrak{a}}) = r_i + \gamma \sum_{s', v'} p(s', v' | s, v, a_{\mathfrak{e}}, a_{\mathfrak{a}}) \tilde{v}_i(s', v'_{\mathfrak{e}}, v'_{\mathfrak{a}}, \pi^*_{\mathfrak{e}}, \pi^*_{\mathfrak{a}})$$

- q_{ij} represents the Q-function for agent i as learned by agent j.
- Each agent learns the Q-functions of **all agents** in the system to calculate current equilibria locally.

Unified Framework

QRM-SG Algorithm Architecture

Actor Mechanics (Action Selection)

- Balances policy exploration using an ϵ-greedy wrapper around calculated stage-game equilibrium solutions.
- With probability $1 - \epsilon$, the model tracks the maximum probability mixed-strategy weights output by the **Lemke-Howson algorithm**.

Critic Updates (Learning Loop)

Values are updated according to standard stochastic approximation:

$$q_{ji} \leftarrow (1 - \alpha_t) q_{ji} + \alpha_t (r_{i,t} + \gamma \overline{q}_{ji}(s', v'_{\epsilon}, v'_{\alpha}))$$

where $\overline{q}_{ji}$ is the expected value scalar at the next augmented state under best-response actions.

QRM-SG Phase 1: Initialization

Process Breakdown (Lines 1–4)

Initialize $\alpha_t, \gamma, \epsilon$

$s \leftarrow \mathit{InitialState}(); v_{\mathfrak{e}} \leftarrow v_{I,\mathfrak{e}}; v_{\mathfrak{a}} \leftarrow v_{I,\mathfrak{a}}$

$q_{\mathfrak{e}i}(\cdot), q_{\mathfrak{a}i}(\cdot) \leftarrow \mathit{InitialQfunctions}()$

Analysis:

- Each agent starts with a "dual-view" Q-table, estimating values for self and opponent to allow local equilibrium computation.
- State is defined in the product space $S \times V_{\mathfrak{e}} \times V_{\mathfrak{a}}$, ensuring the Markov property is preserved.

QRM-SG Phase 2: Nash-Action Selection

Process Breakdown (Lines 7–13)

if $\bar{\epsilon} < \epsilon$ **then**

$a_i \leftarrow \mathit{ChooseActionRandomly}()$

else

$\pi_{\mathfrak{e}i}, \pi_{\mathfrak{a}i} \leftarrow \mathit{CalculateNashEquilibrium}(q_{\mathfrak{e}i}, q_{\mathfrak{a}i})$

$a_i \leftarrow \text{argmax}\ \pi_{ii}(a|s, v_{\mathfrak{e}}, v_{\mathfrak{a}})$

end if

Analysis:

- **Exploration:** Standard ϵ-greedy mechanism balances discovery and optimization.
- **Exploitation:** Agents solve a bimatrix stage game using the **Lemke-Howson method** to find mixed strategy equilibria.

QRM-SG Phase 3: Reward Machine Dynamics

Process Breakdown (Lines 14–18)

$s' \leftarrow \mathit{ExecuteAction}(s, a_{\mathfrak{e}}, a_{\mathfrak{a}})$

$l_t \leftarrow L(s, a_{\mathfrak{e}}, a_{\mathfrak{a}}, s')$

$v_i' \leftarrow \delta_i(v_i, l_t)$; $r_{i,t} \leftarrow \sigma_i(v_i, l_t)$

Analysis:

- Bridges low-level physical dynamics ($s \to s'$) with high-level automata state tracking.
- The labeling function L detects high-level environmental events (e.g., "power base reached").
- Rewards $r_{i,t}$ are generated by the RM transition, making them context and history-aware.

QRM-SG Phase 4: Q-Value Update (Critic)

Process Breakdown (Lines 19–24)

$\pi_{ji}(\cdot|s', v') \leftarrow Nash(q_{ei}(s', v'), q_{ai}(s', v'))$

$\overline{q}_{ji} \leftarrow \pi_{ei}\pi_{ai}q_{ji}(s', v')$

$q_{ji} \leftarrow (1 - \alpha_t)q_{ji} + \alpha_t(r_{j,t} + \gamma\overline{q}_{ji})$

Analysis:

- **Bootstrapping:** Updates current Q-values by computing the Nash equilibrium of the **future** state (s', v').
- **Scalarization:** $\overline{q}_{ji}$ is the expected value of agent j when both follow the Nash strategy profile.
- **Coupled Update:** Agents must learn both Q-functions concurrently to maintain equilibrium local estimates.

Theoretical Foundations

Convergence Assumptions

Assumption 1: Exploration

Every state $s \in S$, RM states $v_{\mathfrak{e}} \in V_{\mathfrak{e}}$, $v_{\mathfrak{a}} \in V_{\mathfrak{a}}$, and actions $a_{\mathfrak{e}} \in A_{\mathfrak{e}}$, $a_{\mathfrak{a}} \in A_{\mathfrak{a}}$, are visited infinitely often when the number of episodes goes to infinity.

Assumption 2: Learning Rate

The learning rate α_t satisfies the following conditions for all t:

1. $0 < \alpha_t < 1$, $\sum_t \alpha_t(s, v_{\mathfrak{e}}, v_{\mathfrak{a}}, a_{\mathfrak{e}}, a_{\mathfrak{a}}) = \infty$, $\sum_t [\alpha_t(s, v_{\mathfrak{e}}, v_{\mathfrak{a}}, a_{\mathfrak{e}}, a_{\mathfrak{a}})]^2 < \infty$, and the latter two hold uniformly and with probability 1.
2. $\alpha_t(s, v_{\mathfrak{e}}, v_{\mathfrak{a}}, a_{\mathfrak{e}}, a_{\mathfrak{a}}) = 0$ if $(s, v_{\mathfrak{e}}, v_{\mathfrak{a}}, a_{\mathfrak{e}}, a_{\mathfrak{a}}) \neq (s^t, v_{\mathfrak{e}}^t, v_{\mathfrak{a}}^t, a_{\mathfrak{e}}^t, a_{\mathfrak{a}}^t)$.

Convergence Assumptions

Assumption 3: Stage Game Structure

One of the following conditions holds during learning:

1. Every stage game $(q_{\mathfrak{e}i}(s, v_{\mathfrak{e}}, v_{\mathfrak{a}}), q_{\mathfrak{a}i}(s, v_{\mathfrak{e}}, v_{\mathfrak{a}}))$, $i \in \{\mathfrak{e}, \mathfrak{a}\}$, for all $t, s, v_{\mathfrak{e}}$, and $v_{\mathfrak{a}}$, has a **global optimal point** $(\tilde{\pi}_{\mathfrak{e}i}, \tilde{\pi}_{\mathfrak{a}i})$ (i.e., $\tilde{\pi}_{ji} q_{ji} \geq \tilde{\pi}'_{ji} q_{ji}$ for any $\tilde{\pi}'_{ji}$, $j \in \{\mathfrak{e}, \mathfrak{a}\}$) and agents' rewards in this equilibrium are used to update their Q-functions.
2. Every stage game $(q_{\mathfrak{e}i}(s, v_{\mathfrak{e}}, v_{\mathfrak{a}}), q_{\mathfrak{a}i}(s, v_{\mathfrak{e}}, v_{\mathfrak{a}}))$, $i \in \{\mathfrak{e}, \mathfrak{a}\}$, for all $t, s, v_{\mathfrak{e}}$, and $v_{\mathfrak{a}}$, has a **saddle point** $(\tilde{\pi}_{\mathfrak{e}i}, \tilde{\pi}_{\mathfrak{a}i})$ (i.e., $\tilde{\pi}_{ji} \tilde{\pi}_{-ji} q_{ji} \geq \tilde{\pi}'_{ji} \tilde{\pi}_{-ji} q_{ji}$ for any $\tilde{\pi}'_{ji}$, $\tilde{\pi}_{ji} \tilde{\pi}_{-ji} q_{ji} \leq \tilde{\pi}_{ji} \tilde{\pi}'_{-ji} q_{ji}$ for any $\tilde{\pi}'_{-ji}$, $j \in \{\mathfrak{e}, \mathfrak{a}\}$), and agents' rewards in this equilibrium are used to update their Q-functions.

Theorem 1: Convergence Guarantee

Theorem 1

Under Assumptions 1, 2 and 3, the sequence $q_{it} = (q^t_{\mathfrak{e}i}, q^t_{\mathfrak{a}i})$ at time t, $i \in \{\mathfrak{e}, \mathfrak{a}\}$ updated by:

$$\begin{aligned} q_{ji}(s, v, a) =& (1 - \alpha_t) q_{ji}(s, v, a) + \\ & \alpha_t (r_{j,t} + \gamma \pi_{\mathfrak{e}i}(\cdot|s', v') \pi_{\mathfrak{a}i}(\cdot|s', v') q_{ji}(s', v')) \end{aligned}$$

for $j \in \{\mathfrak{e}, \mathfrak{a}\}$, where $(\pi_{\mathfrak{e}i}(\cdot|s', v'), \pi_{\mathfrak{a}i}(\cdot|s', v'))$ is the appropriate type of Nash equilibrium solution for the stage game $(q^t_{\mathfrak{e}i}(s', v'), q^t_{\mathfrak{a}i}(s', v'))$, converges to the Q-functions at a Nash equilibrium $(q^*_{\mathfrak{e}i}, q^*_{\mathfrak{a}i})$.

Algorithm 1: QRM-SG Full Pseudocode

Initialize $q_{\mathfrak{e}i}$ and $q_{\mathfrak{a}i}$ for all $i \in \{\mathfrak{e}, \mathfrak{a}\}$
for episode $l = 1, 2, \ldots$ **do**
  $s \leftarrow s_l$, $v_{\mathfrak{e}} \leftarrow v_{l,\mathfrak{e}}$, $v_{\mathfrak{a}} \leftarrow v_{l,\mathfrak{a}}$
  for $t = 0$ **to** *eplength* **do**
    Choose action a_i using ϵ-greedy policy and **Lemke-Howson**
    Execute actions, observe s', and high-level event $l_t = L(s, a_{\mathfrak{e}}, a_{\mathfrak{a}}, s')$
    Update RM states: $v_i' \leftarrow \delta_i(v_i, l_t)$ and reward $r_{i,t} \leftarrow \sigma_i(v_i, l_t)$
    Compute Nash strategy $\pi_{ji}(\cdot|s', v')$ at next augmented state
    Update Q-functions:
    $q_{ji} \leftarrow (1 - \alpha_t) q_{ji} + \alpha_t (r_{j,t} + \gamma \sum \pi\pi q'_{ji})$
    $s \leftarrow s'$, $v_{\mathfrak{e}} \leftarrow v'_{\mathfrak{e}}$, $v_{\mathfrak{a}} \leftarrow v'_{\mathfrak{a}}$
  end for
end for

Algorithm Analysis and Convergence Properties

- **Off-Policy Learning:** RMs enable decomposition of complex RL problems into structured subproblems for efficient learning.
- **Coupled Learning:** The learning processes of agents are coupled because the high-level events depend on joint actions.
- **Decentralized View:** Each agent maintains a "centralized" perspective by learning everyone's Q-functions to compute equilibria locally.

Mathematical Foundation: Convergence Proof

The proof relies on viewing the update as a stochastic approximation of a mapping F^t:

Key Lemmas from Supplementary Materials

- **Lemma 2:** Establishes general convergence for iterations $q^{t+1} = (1 - \alpha_t)q^t + \alpha_t[F^t q^t]$.
- **Lemma 4:** Shows that Q-functions at a Nash equilibrium (q^*) are fixed points of F^t, i.e., $q^* = \mathbb{E}[F^t q^*]$.
- **Lemma 7:** Proves F^t is a **contraction mapping** under Assumption 3:

$$||F^t q - F^t \dot{q}|| \leq \gamma ||q - \dot{q}||$$

Results and Applications

Experimental Framework & Baselines

- **Testing Environment:** Evaluated on 6×6 (and 12×12) competitive, sparse-reward grid world variations of Pac-Man.
- **Comparative Baselines:**
  - `Nash-Q`: Pure spatial state-action tracking.
  - `Nash-QAS`: Incorporates an unstructured history status vector.
  - `MADDPG-SG` & `MADDPG-AS`: Deep multi-agent deterministic policy gradient variants.

Case Study Empirical Analysis

- **Case Study I (Sequential Ordering):** QRM-SG converges to equilibrium by $\approx 7,500$ episodes, whereas unaugmented baselines fail entirely.
- **Case Study II & III (Energy Loops):** Explores complex recharge loops with randomized origin points. QRM-SG reaches stability within $\approx 1,000$ to $1,500$ runs.
- **Scalability:** Successfully finds the Nash equilibrium on an expanded 12×12 map space by episode 21,000.

Conclusion

Summary & Future Outlook

- **Core Contribution:** Bridges reward machines with MARL to model non-Markovian constraints in stochastic games.
- **Robust Architecture:** Augmented state transformations allow tabular actors to navigate complex environments successfully.
- **Future Directions:** Dynamically learning unknown reward machine configurations simultaneously during policy calculation; extensions to general-sum games with more agents.

References

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